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Lab 8: Manage AWS Auto Scaling Groups

Submission: To follow this tutorial. Submission screenshots of your steps and proof of deployment.

https://developer.hashicorp.com/terraform/tutorials/aws/aws-asg

Lab:

Steps :

- **git clone https://github.com/hashicorp-education/learn-terraform-aws-asg**
- **cd learn-terraform-aws-asg**
- **Review configure 'main.tf'**

- terraform init -> terraform applyNext Steps:
  - (1) **Review configuration** (EC2 Launch Template, Auto Scaling group, Load balancer resources, Security groups then "Apply configuration")
  - (2) **Scale instance**s: Use the AWS CLI to scale the number of instances in your ASG

aws autoscaling set-desired-capacity --auto-scaling-group-name $(terraform output -raw asg_name) --desired-capacity 2

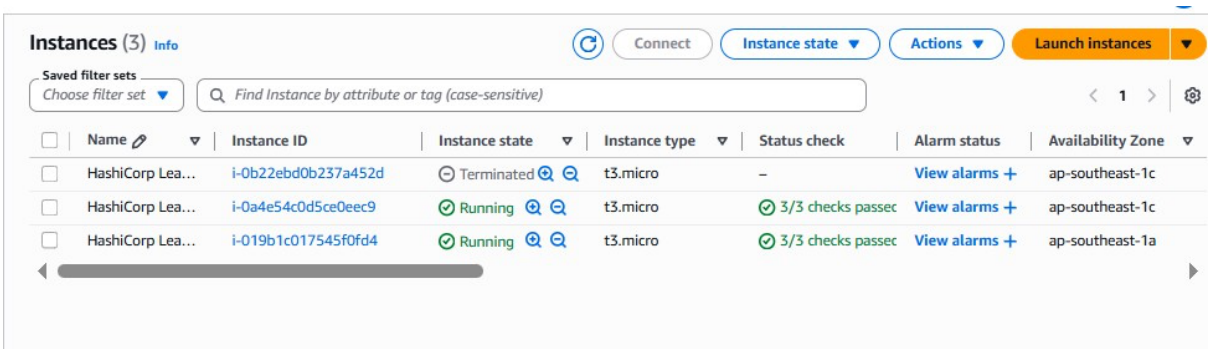
- (3) **Set Life cycle rule**: Add a lifecycle rule to **ignore changes** so Terraform doesn't scale or modify target groups during updates

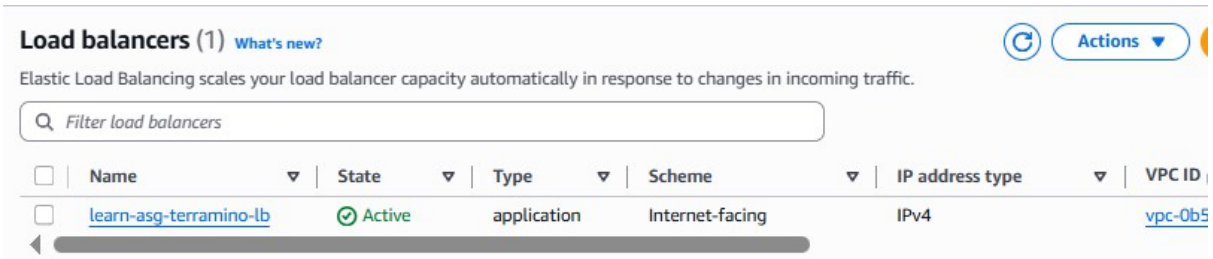
```
lifecycle {
  ignore_changes = [desired_capacity, target_group_arns]
}
```

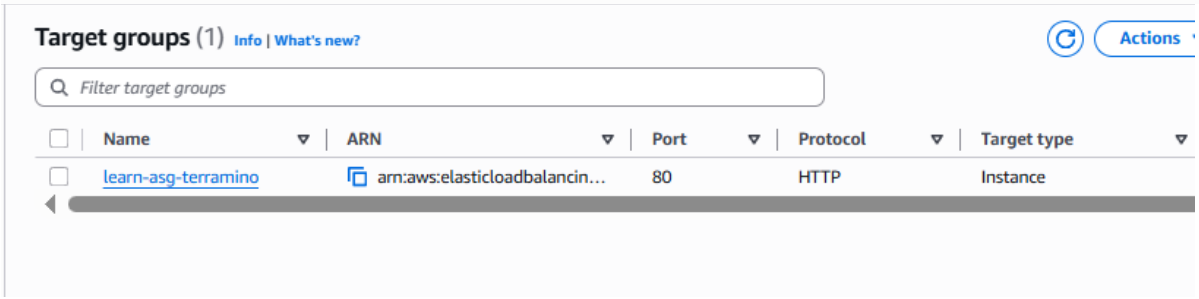
- (4) **Add scaling policy**:

```
Apply complete! Resources: 18 added, 0 changed, 0 destroyed.

Outputs:

application_endpoint = "http://learn-asg-terramino-lb-732640515.ap-southeast-1.elb.amazonaws.com/index.php"
asg_name = "terramino"
lb_endpoint = "http://learn-asg-terramino-lb-732640515.ap-southeast-1.elb.amazonaws.com"
```

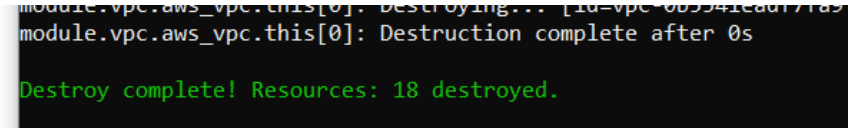

```
{
    "AutoScalingGroups": [
        {
            "AutoScalingGroupName": "terramino",
            "AutoScalingGroupARN": "arn:aws:autoscaling:ap-southeast-1:804329958785:autoScalingGroup:1bb25c32-1b67-4cc2-
9f9b-75f1e692fea6:autoScalingGroupName/terramino",
            "LaunchTemplate": {
                "LaunchTemplateId": "lt-01f801366fd145c6c",
                "LaunchTemplateName": "learn-terraform-aws-asg-20260327065628039000000001",
                "Version": "$Latest"
            },
            "MinSize": 1,
            "MaxSize": 3,
            "DesiredCapacity": 2,
            "DefaultCooldown": 300,
            "AvailabilityZones": [
                "ap-southeast-1a",
                "ap-southeast-1c",
                "ap-southeast-1b"
            ],
            "LoadBalancerNames": [],
            "TargetGroupARNs": [
                "arn:aws:elasticloadbalancing:ap-southeast-1:804329958785:targetgroup/learn-asg-terramino/0455cf7dc1c077
10"
            ],
            "HealthCheckType": "ELB",
            "HealthCheckGracePeriod": 300,
            "Instances": [
```

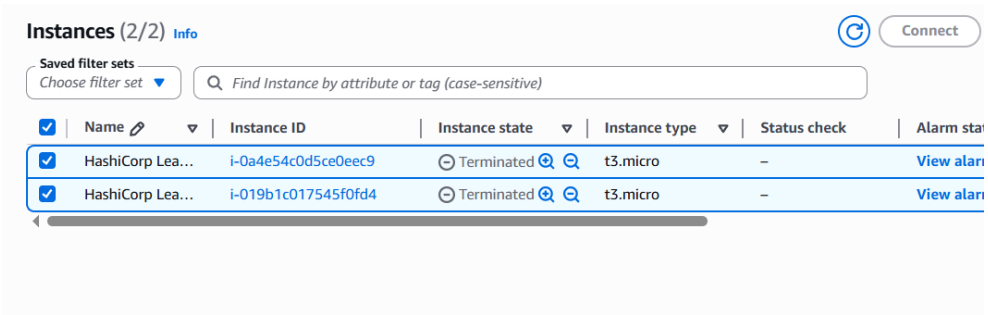






- "**Destroy Configuration**" | $ terraform destroy





Questions:

1. What is a launch configuration or launch template?

2. Why do we use a Load Balancer with ASG?

3. Why doesn't Terraform directly manage individual EC2 instances in ASG?

4. Explain state drift in Terraform.

Answers:

2. Launch Configuration / Launch Template

- A blueprint for EC2 in an ASG, Specifies AMI, instance type, security group, user data.

- Launch Template = modern, versioned, flexible.

3. It spreads traffic across all the running instances. Keeps the app **available** and **fast**, and stops traffic going to broken instances.

4. We use a Load Balancer with ASG because ASG handles instances automatically (scaling, replacing). Terraform only manages the **ASG itself**, not each instance, to avoid conflicts.

5. State Drift happens when the **real AWS setup changes outside Terraform**. Terraform's record and reality don't match anymore. Always use Terraform to make changes to avoid this.